Day 01

FULL STACK FUNDAMENTALS

Python Full Stack Development Internship

Topic: FULL STACK FUNDAMENTALS

Prepared By: - varsha Purushottam mhaske

Date: - 13 October 2026

1.What Is Full Stack Development?

full Stack Development is the process of building a web application by working with both the parts users see and the parts that operate behind the scenes. A full stack application commonly includes a frontend, a backend, and a database.

Example: In an online shopping website, the frontend displays products, t backend processes orders, and the database stores product, customer, and order information.

2.frontend, backend database?

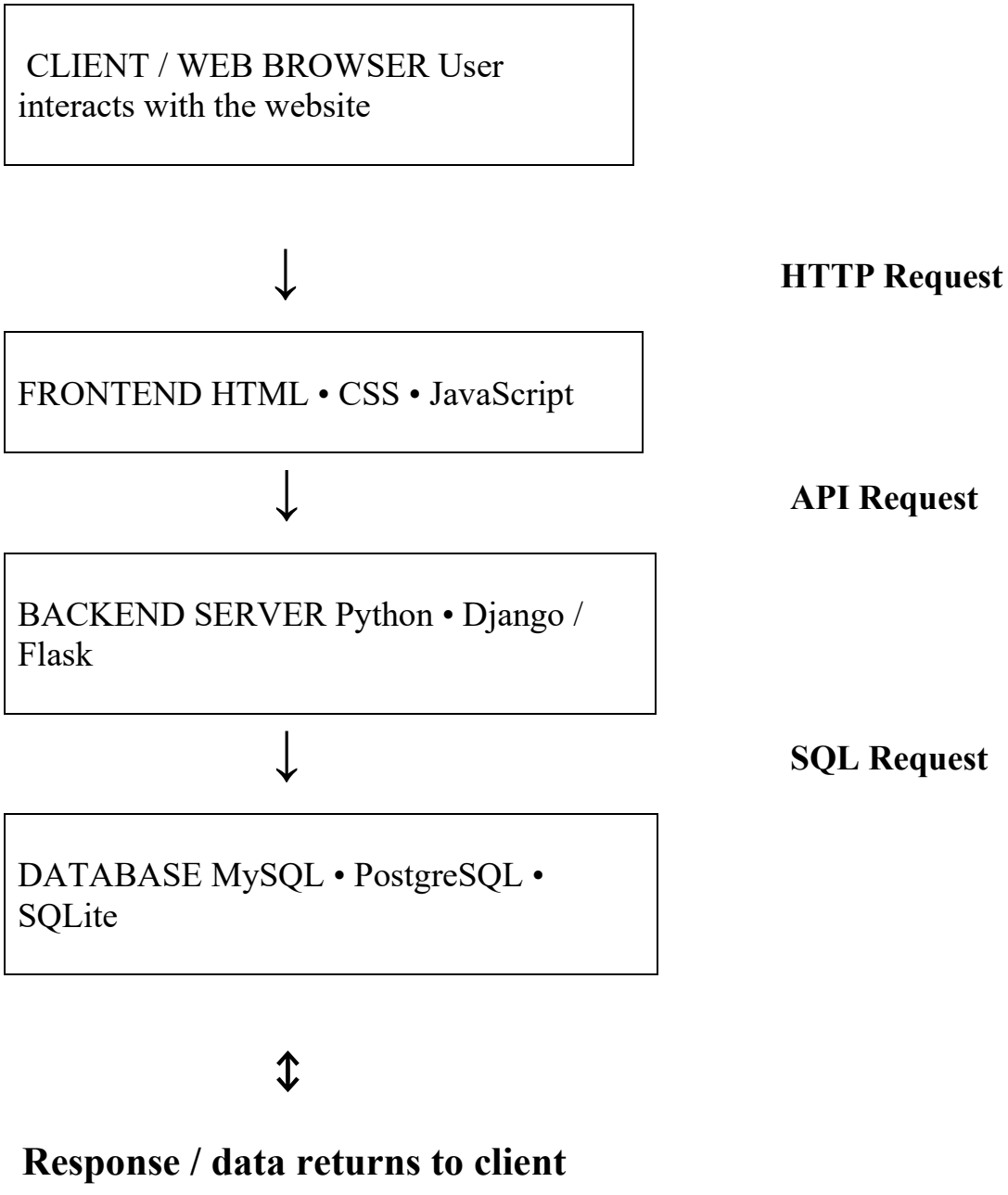
Component	Role	Common technologies
Frontend	The user interface: pages, buttons, forms, and content that users see and interact with.	HTML, CSS, JavaScript
Backend	Processes requests, applies business rules, handles authentication, and communicates with the database.	Python, Django, Flask
Database	Stores, organizes, updates, and retrieves application data.	MySQL, PostgreSQL, SQLite

3.Client–Server Architecture?

Python, Django, Flask MySQL, PostgreSQL, SQLite Client–Server Architecture is a model in which a client (such as a web browser) sends a request to a server. The server processes the request and returns a response. If the request needs stored information, the server can communicate with a database.

Main parts: Client — sends requests; frontend — presents the interface; backend server — runs application logic; database — stores and returns data.

Architecture Diagram:



4. HTTP Request–Response Flow?

HTTP (Hypertext Transfer Protocol) is used for communication between a client and a web server. A typical request–response cycle works as follows:

1. The user performs an action, such as opening a page or clicking a button.

2. The browser sends an HTTP request to the backend server.

3. The server checks the request and runs the required application logic.

4. If needed, the backend queries the database.

5. The database returns the requested data to the backend.

6. The server sends an HTTP response, including a status code and often data or HTML.

7. The browser uses the response to display or update the page.

Common HTTP methods:

Method	Purpose	Example
• GET	Retrieve data	Get the list of students
• POST	Submit data or create a resource	Add a new student
• PUT	Replace or update a resource	Update student details
• DELETE	Delete a resource	Delete student record

5.what is a REST API?

REST stands for Representational State Transfer. An API (Application Programming Interface) allows different software applications to communicate with each other.

A REST API allows clients to interact with server resources using HTTP methods.

Example: Student Management System

HTTP Method	Endpoint	Purpose
GET	/Students	Retrieve all students
GET	/students/1	Retrieve a particular student
POST	/Students	Create a student

PUT	`/students/1`	Update a student
DELETE	`/students/1`	Delete a student

6. Python full stack technology stack:

Layer	Technologies
Frontend	HTML, CSS, JavaScript
Backend	Python
Backend Framework	Django or Flask
Database	MySQL, PostgreSQL, SQLite
API	REST API
Tools	VS Code, Git, GitHub
Testing	Browser testing,

7. Real-world application example:

Example: Online Shopping Application

Consider an online shopping website.

1. The user opens the website and views products.
2. The frontend displays product details.
3. The user adds a product to the cart.
4. The backend processes the request.
5. The backend retrieves product details and prices from the database.
6. The server returns the response.
7. The frontend displays the updated cart